Exercise J - A

1. For $x \in (0, \pi)$ the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has : **[JEE(Adv.)2014]**

(A) infinitely many solutions (B) three solutions

(C) one solution (D) no solution

$x \in (0, \pi)$ के लिए समीकरण $\sin x + 2\sin 2x - \sin 3x = 3$ के : **[JEE(Adv.)2014]**

(A) अनन्त हल होगें (B) तीन हल होगें

(C) एक हल होगा (D) कोई हल नहीं होगा

Ans. D

Sol. $\sin x + 2\sin 2x - \sin 3x = 3.$

$\Rightarrow \sin x\,(1 + 2\cos x - 3 + 4\sin^2 x) = 3.$

$\Rightarrow (4\sin^2 x + 2\cos x - 2) = \dfrac{3}{\sin x}$

$\Rightarrow 2 - 4\cos^2 x + 2\cos x = \dfrac{3}{\sin x}$

$\Rightarrow \dfrac{9}{4} - \left(2\cos x - \dfrac{1}{2}\right)^2 = \dfrac{3}{\sin x}.$

L.H.S. $\le \dfrac{9}{4}$ R.H.S. $\ge 3.$

No solution

2. The numer of distinct solutions of the equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is. **[JEE(Adv.)2015]**

अंतराल $[0, 2\pi]$ में समीकरण $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ के विभिन्न हलों की संख्या है।

[JEE(Adv.)2015]

Ans. 8

Sol. $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$

$\Rightarrow \quad \frac{5}{4}\cos^2 2x + 1 - \frac{1}{2}\sin^2 2x + 1 - \frac{3}{4}\sin^2 2x = 2$

$\Rightarrow \quad \cos^2 2x = \sin^2 2x$

$\Rightarrow \quad \tan^2 2x = 1$

$\because \quad 2x \in [0, 4\pi] \quad \Rightarrow \quad x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$

so number of solution is 8

3. Let $S=\left\{x\in(-\pi,\pi): x\neq 0,\pm\frac{\pi}{2}\right\}$. The sum of all distinct solutions of the equation $\sqrt{3}\sec x+\operatorname{cosec} x+2(\tan x-\cot x)=0$ in the set S is equal to : **[JEE(Adv.)2016]**

माना कि $S=\left\{x\in(-\pi,\pi): x\neq 0,\pm\frac{\pi}{2}\right\}$ है। समुच्चय S में समीकरण $\sqrt{3}\sec x+\operatorname{cosec} x+2(\tan x-\cot x)=0$ के सभी भिन्न हलों का योग है : **[JEE(Adv.)2016]**

(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

Ans. C

Sol. $2\left(\frac{\sqrt{3}/2}{\cos x}+\frac{1/2}{\sin x}\right)+2(\tan x-\cot x)=0$

$\Rightarrow 2\frac{\cos(x-60°)}{\sin x\cos x}+\frac{2(-\cos 2x)}{\sin x\cos x}=0$

$\Rightarrow \cos(x-60°)=\cos 2x$

$\Rightarrow x=2n\pi-\frac{\pi}{3},\ (6m+1)\frac{\pi}{9}\ (m, n\in I)$

$\Rightarrow$ Solutions belonging to set $S=-\frac{\pi}{3},-\frac{5\pi}{9},\frac{\pi}{9},\frac{7\pi}{9}$

sum = 0

4. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x+2b\sin x=c, x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha+\beta=\pi/3$. Then, the value of b/a is _____. **[JEE(Adv.)2018]**

माना कि a, b, c ऐसी तीन शून्येत्तर वास्तविक संख्याऐं हैं जिनके लिये समीकरण $\sqrt{3}a\cos x+2b\sin x=c, x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, के दो भिन्न वास्तविक मूल α और β हैं, जहाँ $\alpha+\beta=\pi/3$. तब b/a का मान है। **[JEE(Adv.)2018]**

Ans. 0.5

Sol. $\sqrt{3}\ a\cos x+2b\sin x=c \quad x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$\Rightarrow \sqrt{3}\ a\left(\frac{1-t^2}{1+t^2}\right)+2b\left(\frac{2t}{1+t^2}\right)=c,$ where $t=\tan\frac{x}{2}$

$\Rightarrow \sqrt{3}\ a(1-t^2)+4bt=c(1+t^2)$

$$\Rightarrow t^2(c+\sqrt{3}\,a)-4\,bt+c-\sqrt{3}\,a=0$$

$$\because \ \frac{\alpha+\beta}{2}=\frac{\pi}{6}$$

$$\Rightarrow \tan\left(\frac{\alpha+\beta}{2}\right)=\frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{t_1+t_2}{1-t_1t_2}=\frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{4b}{c+\sqrt{3}\,a-c+\sqrt{3}\,a}=\frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{b}{a}=\frac{1}{2}$$

5. Consider the following column : **[JEE(Adv.)2022]**

Column - I	**Column - II**
(I) $\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]:\cos x+\sin x=1\right\}$	(P) has two elements
(II) $\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]:\sqrt{3}\tan 3x=1\right\}$	(Q) has three elements
(III) $\left\{x\in\left[-\frac{6\pi}{5},\frac{6\pi}{5}\right]:2\cos(2x)=\sqrt{3}\right\}$	(R) has four elements
(IV) $\left\{x\in\left[-\frac{7\pi}{4},\frac{7\pi}{4}\right]:\sin x-\cos x=1\right\}$	(S) has five elements
	(T) has six elements

The correct option is:

(A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)

(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

निम्न कॉलम पर विचार कीजिए : **[JEE(Adv.)2022]**

Column - I	**Column - II**
(I) $\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]:\cos x+\sin x=1\right\}$	(P) में दो अवयव है
(II) $\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]:\sqrt{3}\tan 3x=1\right\}$	(Q) में तीन अवयव है

(III) $\left\{x \in\left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right]: 2\cos(2x)=\sqrt{3}\right\}$ (R) में चार अवयव है

(IV) $\left\{x \in\left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right]: \sin x-\cos x=1\right\}$ (S) में पांच अवयव है

(T) में छः अवयव है

सही विकल्प है :

(A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)
(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

Ans. (B)

Sol. $\left\{x \in\left[\frac{-2\pi}{3}, \frac{2\pi}{3}\right], \cos x+\sin x=1\right\}$

$\cos x+\sin x=1$

$$\sin\left(\frac{\pi}{4}+x\right)=\frac{1}{\sqrt{2}}$$

$$\frac{\pi}{4}+x=n\pi+(-1)^n\frac{\pi}{4}$$

$$x=n\pi+(-1)^n\frac{\pi}{4}-\frac{\pi}{4}$$

$\therefore$ x has 2 elements. $\rightarrow$ P

(ii) $\left\{x \in\left[\frac{-5\pi}{18}, \frac{5\pi}{18}\right]: \sqrt{3}\tan 3x=1\right\}$

$$\sqrt{3}\tan 3x=1$$

$$\tan 3x=\frac{1}{\sqrt{3}}$$

$$3x=n\pi+\frac{\pi}{6}$$

$$x=\frac{n\pi}{3}+\frac{\pi}{18}$$

$\therefore$ x has 2 elements. $\rightarrow$ P

(iii) $\left\{x \in\left[\frac{-6\pi}{5}, \frac{6\pi}{5}\right]: 2\cos 2x=\sqrt{3}\right\}$

$$2\cos 2x=\sqrt{3}$$

$\cos 2x = \frac{\sqrt{3}}{2}$

$2x = 2n\pi \pm \frac{\pi}{6}$

$x = n\pi \pm \frac{\pi}{12}$

$\therefore$ x has 6 elements. $\rightarrow T$

(iv) $\left\{x \in \left[\frac{-7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$

$\sin x - \cos x = 1$

$\sin\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$x - \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$

$x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}$

$\therefore$ x has 4 elements. $\rightarrow R$

$\therefore$ option B is correct.